

Classified as hazardous in accordance with the criteria of EPA New Zealand

## Section 1 - Identification

### Product Identifier

|                             |                                              |
|-----------------------------|----------------------------------------------|
| <b>Product Name</b>         | <b><u>Mercury(II) bromide, ultra dry</u></b> |
| <b>CAS No</b>               | 7789-47-1                                    |
| <b>Synonyms</b>             | Mercuric bromide                             |
| <b>Molecular Formula</b>    | Br <sub>2</sub> Hg                           |
| <b>Molecular Weight</b>     | 360.4                                        |
| <b>Recommended Use</b>      | Laboratory chemicals.                        |
| <b>Uses advised against</b> | No Information available                     |

|                                |                                                                                             |
|--------------------------------|---------------------------------------------------------------------------------------------|
| <b>Product Code</b>            | <b>35711</b>                                                                                |
| <b>Address</b>                 | Thermo Fisher Scientific New Zealand Ltd<br>244 Bush Road, Albany,<br>Auckland, New Zealand |
| <b>Emergency Tel.</b>          | <b>CHEMTREC®</b><br><b>09 980 6780 or +64 9 980 6780</b>                                    |
| <b>Telephone / Fax Numbers</b> | Tel: 09 980 6700<br>Fax: 09 980 6788                                                        |
| <b>E-mail address</b>          | <a href="mailto:ANZinfo@thermofisher.com">ANZinfo@thermofisher.com</a>                      |

## Section 2 - Hazard(s) Identification

### Classification under Work Safe New Zealand

Classified as hazardous in accordance with the criteria of EPA New Zealand

**HSNO Approval Number      HSR004544**

### GHS Classification

#### Physical hazards

Based on available data, the classification criteria are not met

#### Health hazards

|                                                      |            |
|------------------------------------------------------|------------|
| Acute Oral Toxicity                                  | Category 2 |
| Acute Dermal Toxicity                                | Category 1 |
| Acute Inhalation Toxicity - Dusts and Mists          | Category 2 |
| Specific target organ toxicity - (repeated exposure) | Category 2 |

#### Environmental hazards

|                          |            |
|--------------------------|------------|
| Acute aquatic toxicity   | Category 1 |
| Chronic aquatic toxicity | Category 1 |

Label Elements

Signal Word

Danger

**Hazard Statements**

H373 - May cause damage to organs through prolonged or repeated exposure

H410 - Very toxic to aquatic life with long lasting effects

H300 + H310 + H330 - Fatal if swallowed, in contact with skin or if inhaled

**Precautionary Statements****Prevention**

P262 - Do not get in eyes, on skin, or on clothing

P264 - Wash face, hands and any exposed skin thoroughly after handling

P270 - Do not eat, drink or smoke when using this product

P271 - Use only outdoors or in a well-ventilated area

P280 - Wear protective gloves/protective clothing/eye protection/face protection

P273 - Avoid release to the environment

**Response**

P302 + P352 - IF ON SKIN: Wash with plenty of soap and water

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing

P310 - Immediately call a POISON CENTER or doctor

P330 - Rinse mouth

P361 + P364 - Take off immediately all contaminated clothing and wash it before reuse

P391 - Collect spillage

**Storage**

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

P405 - Store locked up

**Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

**Other hazards which do not result in classification**

Toxic to terrestrial vertebrates

## Section 3 - Composition and Information on Ingredients

| Component        | CAS No    | Weight % |
|------------------|-----------|----------|
| Mercuric bromide | 7789-47-1 | >95      |

## Section 4 - First Aid Measures

Description of first aid measures**New Zealand Emergency Tel.**CHEMTREC®  
09 980 6780 or +64 9 980 6780**Inhalation**

Remove to fresh air. Immediate medical attention is required. If not breathing, give artificial respiration.

**Eye Contact**

Immediate medical attention is required. Rinse immediately with plenty of water, also under

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|                                            |                                                                                                                                                  |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
|                                            | the eyelids, for at least 15 minutes.                                                                                                            |
| <b>Skin Contact</b>                        | Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.                                      |
| <b>Ingestion</b>                           | Do NOT induce vomiting. Call a physician immediately.                                                                                            |
| <b>Self-Protection of the First Aider</b>  | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination. |
| <b>First Aid Facilities</b>                | Eyewash, safety shower and washroom.                                                                                                             |
| <b>Most important symptoms and effects</b> | No information available.                                                                                                                        |
| <b>Notes to Physician</b>                  | Treat symptomatically.                                                                                                                           |

## **Section 5 - Fire Fighting Measures**

### **Suitable Extinguishing Media**

Water spray. Carbon dioxide (CO<sub>2</sub>). Dry chemical. Chemical foam.

### **Extinguishing media which must not be used for safety reasons**

No information available.

### **Specific Hazards Arising from the Chemical**

Do not allow run-off from fire-fighting to enter drains or water courses.

### **Hazardous Combustion Products**

Hydrogen halides.

### **Special protective equipment and precautions for fire fighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## **Section 6 - Accidental Release Measures**

### **Personal Precautions, Protective Equipment and Emergency Procedures**

#### **Emergency procedures**

Ensure adequate ventilation.

#### **Environmental Precautions**

Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained.

#### **Methods for Containment and Clean Up**

Wear self-contained breathing apparatus and protective suit. Sweep up and shovel into suitable containers for disposal. Do not let this chemical enter the environment.

#### **Precautions to prevent secondary hazards**

Clean contaminated objects and areas thoroughly observing environmental regulations

#### **Reference to Other Sections**

Refer to protective measures listed in Sections 8 and 13.

## **Section 7 - Handling and Storage**

### **Precautions for Safe Handling**

**Advice on safe handling**

Ensure adequate ventilation. Wear personal protective equipment/face protection. Do not breathe dust. Do not get in eyes, on skin, or on clothing. Do not ingest. If swallowed then seek immediate medical assistance. Avoid dust formation.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

**Conditions for Safe Storage, Including any Incompatibilities****Storage Conditions**

Keep in a dry, cool and well-ventilated place. Keep container tightly closed. Keep locked up.

**Incompatible Materials**

Strong oxidizing agents.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

## Section 8 - Exposure Controls and Personal Protection

**Control parameters****Exposure limits**

**AUS** - Exposure Standards for Atmospheric Contaminants in the Occupational Environment - Guidance Note on the Interpretation of Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:3008(1995)]

Adopted National Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:1003(1995)] updated in August, 2005. Safe Work Australia

**ACGIH** - Threshold Limit Values - Ceiling (TLV-C) guidelines by the American Conference of Governmental Industrial Hygienists (ACGIH) for controlling worker exposure to airborne chemical concentrations in the workplace.

**UK** - EH40/2005 Work Exposure Limits, Fourth edition. Published 2020.

| Component        | New Zealand WEL | Australia                                      | ACGIH TLV                            | The United Kingdom                                                      |
|------------------|-----------------|------------------------------------------------|--------------------------------------|-------------------------------------------------------------------------|
| Mercuric bromide |                 | TWA: 0.003 ppm<br>TWA: 0.025 mg/m <sup>3</sup> | TWA: 0.025 mg/m <sup>3</sup><br>Skin | STEL: 0.06 mg/m <sup>3</sup> 15 min<br>TWA: 0.02 mg/m <sup>3</sup> 8 hr |

**Biological limit values**

**ACGIH** - American Conference of Governmental Industrial Hygienists (ACGIH) TLVs® and BEIs®- Threshold Limit Values for Chemical Substances and Physical Agents & Biological Exposure Indices. 2022 Edition

| Component        | New Zealand | Australia | ACGIH - Biological Exposure Indices                                                                                                                                                                              | United Kingdom |
|------------------|-------------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| Mercuric bromide |             |           | 35 µg/g creatinine<br>Medium: urine<br>Time: prior to shift<br>Determinant: Total inorganic mercury<br>15 µg/L<br>Medium: blood<br>Time: end of shift at end of workweek<br>Determinant: Total inorganic mercury |                |

**Appropriate engineering controls****Engineering Measures**

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

**Individual protection measures, such as personal protective equipment****Eye Protection**

Goggles (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial

applications)

**Hand Protection**

Protective gloves

| Glove material                                 | Breakthrough time                 | Glove thickness | AUS/NZ Standard | Glove comments        |
|------------------------------------------------|-----------------------------------|-----------------|-----------------|-----------------------|
| Natural rubber, Nitrile rubber, Neoprene, PVC. | See manufacturers recommendations | -               | AS/NZS 2161     | (minimum requirement) |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection**

Wear appropriate protective gloves and clothing to prevent skin exposure

**Respiratory Protection**

Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use and maintenance of respiratory protective devices

**Recommended Filter type:  
Recommended half mask:-**

Particulates filter conforming to EN 143 (or AUS/NZ equivalent)  
 Particle filtering: EN149:2001 (or AUS/NZ equivalent)  
 When RPE is used a face piece Fit Test should be conducted

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls**

Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.

## Section 9 - Physical and Chemical Properties

### Information on basic physical and chemical properties

|                                                |                          |                                          |
|------------------------------------------------|--------------------------|------------------------------------------|
| <b>Physical State</b>                          | Powder Solid             |                                          |
| <b>Appearance</b>                              | White                    |                                          |
| <b>Odor</b>                                    | Odorless                 |                                          |
| <b>Odor Threshold</b>                          | No data available        |                                          |
| <b>pH</b>                                      | No information available |                                          |
| <b>Melting Point/Range</b>                     | 236 °C / 456.8 °F        |                                          |
| <b>Softening Point</b>                         | No data available        |                                          |
| <b>Boiling Point/Range</b>                     | 322 °C / 611.6 °F        |                                          |
| <b>Flammability (liquid)</b>                   | Not applicable           | Solid                                    |
| <b>Flammability (solid,gas)</b>                | No information available |                                          |
| <b>Explosion Limits</b>                        | No data available        |                                          |
| <b>Flash Point</b>                             | No information available | <b>Method -</b> No information available |
| <b>Autoignition Temperature</b>                | No data available        |                                          |
| <b>Decomposition Temperature</b>               | No data available        |                                          |
| <b>Viscosity</b>                               | Not applicable           | Solid                                    |
| <b>Water Solubility</b>                        | 0.6g/100ml (25 C)        |                                          |
| <b>Solubility in other solvents</b>            | No information available |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                          |                                          |
| <b>Vapor Pressure</b>                          | No data available        |                                          |
| <b>Density / Specific Gravity</b>              | 6.1090                   |                                          |
| <b>Bulk Density</b>                            | No data available        |                                          |
| <b>Vapor Density</b>                           | Not applicable           | Solid                                    |
| <b>Particle characteristics</b>                | No data available        |                                          |

**Other information**

|                   |                        |
|-------------------|------------------------|
| Molecular Formula | Br <sub>2</sub> Hg     |
| Molecular Weight  | 360.4                  |
| Evaporation Rate  | Not applicable - Solid |

## Section 10 - Stability and Reactivity

|                                  |                                            |
|----------------------------------|--------------------------------------------|
| Reactivity                       | None known, based on information available |
| Stability                        | Light sensitive.                           |
| Sensitivity to Mechanical Impact | No information available                   |
| Sensitivity to Static Discharge  | No information available                   |
| Hazardous Polymerization         | No information available.                  |
| Hazardous Reactions              | No information available.                  |
| Conditions to Avoid              | Exposure to light, Incompatible products.  |
| Incompatible Materials           | Strong oxidizing agents.                   |
| Hazardous Decomposition Products | Hydrogen halides.                          |

## Section 11 - Toxicological Information

### Acute Effects

### Information on likely routes of exposure

#### Product Information

|            |                                                          |
|------------|----------------------------------------------------------|
| Inhalation | Toxic by inhalation. Avoid breathing dust or spray mist. |
| Eyes       | Avoid contact with eyes.                                 |
| Skin       | Avoid contact with skin. Harmful in contact with skin.   |
| Ingestion  | May be harmful if swallowed.                             |

### Numerical measures of toxicity

|                     |            |
|---------------------|------------|
| (a) acute toxicity; |            |
| Oral                | Category 2 |
| Dermal              | Category 1 |
| Inhalation          | Category 2 |

| Component        | LD50 Oral               | LD50 Dermal | LC50 Inhalation |
|------------------|-------------------------|-------------|-----------------|
| Mercuric bromide | LD50 = 40 mg/kg ( Rat ) |             |                 |

(b) skin corrosion/irritation; No data available

(c) serious eye damage/irritation; No data available

(d) respiratory or skin sensitization;  
Respiratory No data available  
Skin No data available

(e) germ cell mutagenicity; No data available

|                             |                                                                                |
|-----------------------------|--------------------------------------------------------------------------------|
| (f) carcinogenicity;        | No data available<br>There are no known carcinogenic chemicals in this product |
| (g) reproductive toxicity;  | No data available                                                              |
| (h) STOT-single exposure;   | No data available                                                              |
| (i) STOT-repeated exposure; | Category 2                                                                     |
| Target Organs               | Kidney, Central nervous system (CNS).                                          |
| (j) aspiration hazard;      | Not applicable<br>Solid                                                        |

**Symptoms / effects, both acute and delayed**  
No information available.

## Section 12 - Ecological Information

### Ecotoxicity

|                                |                                                                                                                                                                                     |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Aquatic ecotoxicity</b>     | Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment. |
| <b>Terrestrial ecotoxicity</b> | There is no data for this product                                                                                                                                                   |

### **Persistence and Degradability**

|                                              |                                                                                                                                                             |
|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Persistence</b>                           | Soluble in water, Persistence is unlikely, based on information available.                                                                                  |
| <b>Degradability</b>                         | Not relevant for inorganic substances.                                                                                                                      |
| <b>Degradation in sewage treatment plant</b> | Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.                                             |
| <b>Bioaccumulative Potential</b>             | Bioaccumulation is unlikely                                                                                                                                 |
| <b>Mobility</b>                              | The product is water soluble, and may spread in water systems. Will likely be mobile in the environment due to its water solubility. Highly mobile in soils |

### Other adverse effects

|                                        |                                                                           |
|----------------------------------------|---------------------------------------------------------------------------|
| <b>Endocrine Disruptor Information</b> | This product does not contain any known or suspected endocrine disruptors |
| <b>Persistent Organic Pollutant</b>    | This product does not contain any known or suspected substance            |
| <b>Ozone Depletion Potential</b>       | This product does not contain any known or suspected substance            |

## Section 13 - Disposal Considerations

**Waste treatment methods****Waste from Residues/Unused Products**

Do not allow into drains or watercourses or dispose of where ground or surface waters may be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations.

**Contaminated Packaging**

Dispose of this container to hazardous or special waste collection point.

**Other Information**

Disposal agencies or waste contractors must comply with the New Zealand Hazardous Substances (Disposal) Regulations. Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Do not let this chemical enter the environment. Do not empty into drains.

## Section 14 - Transport Information

**NZS 5433:2020**

**UN-No** UN1634  
**Proper Shipping Name** MERCURY BROMIDES  
**Hazard Class** 6.1  
**Packing Group** II

**IATA**

**UN-No** UN1634  
**Proper Shipping Name** MERCURY BROMIDES  
**Hazard Class** 6.1  
**Packing Group** II

**IMDG/IMO**

**UN-No** UN1634  
**Proper Shipping Name** MERCURY BROMIDES  
**Hazard Class** 6.1  
**Packing Group** II

| Component                             | IMDG Marine Pollutant                                                                                                                   |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Mercuric bromide<br>7789-47-1 ( >95 ) | IMDG regulated marine pollutant (Listed in the index) IMDG regulated marine pollutant (UN2025) IMDG regulated marine pollutant (UN1634) |

**Environmental hazards**

Dangerous for the environment  
Product is a marine pollutant according to the criteria set by IMDG/IMO

**Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code**

Not applicable, packaged goods

**Special Precautions**

No special precautions required. Please refer to the applicable dangerous goods regulations for additional information.

**Additional information**

None known

## Section 15 - Regulatory Information

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

|                             |           |
|-----------------------------|-----------|
| <b>HSNO Approval Number</b> | HSR004544 |
|-----------------------------|-----------|

**National Regulations**

There are no applicable tolerable exposure limits or environmental exposure limits according to the EPA Controls for Hazardous



## Substances

**Certified handlers, tracking and controlled substance license requirements**

Certified handlers are required for some substances. This includes substances requiring a controlled substance license, and most explosives, vertebrates toxic agents, and certain fumigants. Acutely toxic substances which are a Category 1 or 2, such as pesticides also require Certified handlers. Please check the Health and Safety at Work Act 2015 for further information. Tracking is required for some highly hazardous substances. These substances need to be under the control of an appropriately trained person or appropriately secured. Please check the Health and Safety at Work Act 2015 for further information.

**Prohibition or notification/licensing requirements**

Shown below are details of specific prohibition/notifications or licensing requirements when they apply.

**International Regulations**

**Ozone Depletion Potential** This product does not contain any known or suspected substance

**Persistent Organic Pollutant** This product does not contain any known or suspected substance

**Rotterdam Convention (PIC)** Chemicals Subject to Prior Informed Consent (PIC)

| Component                    | Rotterdam Convention (PIC) |
|------------------------------|----------------------------|
| Mercuric bromide - 7789-47-1 | X                          |

| Component        | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | IMDG Marine Pollutant                                                                                                                   |
|------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Mercuric bromide |                                                                                           |                                                                                          | IMDG regulated marine pollutant (Listed in the index) IMDG regulated marine pollutant (UN2025) IMDG regulated marine pollutant (UN1634) |

**Authorisation/Restrictions according to EU REACH**

| Component        | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances                                                      | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|------------------|---------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Mercuric bromide | -                                                                   | Use restricted. See item 18. (see link for restriction details)<br>Use restricted. See item 75. (see link for restriction details) | -                                                                                                     |

<https://echa.europa.eu/substances-restricted-under-reach>

**International Inventories**

New Zealand (NZIoC), Australia (AICS), Europe (EINECS/ELINCS/NLP), Korea (KECL), China (IECSC), Taiwan (TCSI), Japan (ENCS), Japan (ISHL), Canada (DSL/NDL), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component        | CAS No    | NZIoC | AICS | EINECS | ELINCS | NLP | KECL     | IECSC | TCSI |
|------------------|-----------|-------|------|--------|--------|-----|----------|-------|------|
| Mercuric bromide | 7789-47-1 | X     | X    | -      | -      | -   | KE-23120 | X     | X    |

| Component        | CAS No    | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDL | PICCS | ISHL | ENCS |
|------------------|-----------|------|-----------------------------------------------|-----|-----|-------|------|------|
| Mercuric bromide | 7789-47-1 | X    | ACTIVE                                        | X   | -   | X     | X    | -    |

**Legend:** X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

**Section 16 - Other Information**

**This safety data sheet complies with the requirements of the EPA Hazardous Substances (Hazard Classification) Notice 2020 and WorkSafe New Zealand Regulations**

#### Legend

|                                                                                                      |                                                                                                                              |
|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>NZIoC</b> - New Zealand Inventory of Chemicals                                                    | <b>AICS</b> - Australian Inventory of Chemical Substances                                                                    |
| <b>TSCA</b> - United States Toxic Substances Control Act Section 8(b) Inventory                      | <b>EINECS/ELINCS</b> - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances |
| <b>DSL/NDL</b> - Canadian Domestic Substances List/Non-Domestic Substances List                      | <b>ENCS</b> - Japanese Existing and New Chemical Substances                                                                  |
| <b>IECSC</b> - Chinese Inventory of Existing Chemical Substances                                     | <b>KECL</b> - Korean Existing and Evaluated Chemical Substances                                                              |
| <b>PICCS</b> - Philippines Inventory of Chemicals and Chemical Substances                            | <b>CAS</b> - Chemical Abstracts Service                                                                                      |
| <b>TWA</b> - Time Weighted Average                                                                   | <b>ACGIH</b> - American Conference of Governmental Industrial Hygienists                                                     |
| <b>IARC</b> - International Agency for Research on Cancer                                            | <b>PNEC</b> - Predicted No Effect Concentration                                                                              |
| <b>NZS 5433:2020</b> - Transport of Dangerous Goods on Land                                          | <b>OECD</b> - Organisation for Economic Co-operation and Development                                                         |
| <b>ICAO/IATA</b> - International Civil Aviation Organization/International Air Transport Association | <b>IMO/IMDG</b> - International Maritime Organization/International Maritime Dangerous Goods Code                            |
| <b>MARPOL</b> - International Convention for the Prevention of Pollution from Ships                  | <b>ADG</b> - Australian Code for the Transport of Dangerous Goods by Road and Rail                                           |
| <b>LD50</b> - Lethal Dose 50%                                                                        | <b>LC50</b> - Lethal Concentration 50%                                                                                       |
| <b>EC50</b> - Effective Concentration 50%                                                            | <b>ATE</b> - Acute Toxicity Estimate                                                                                         |
| <b>WEL</b> - Workplace Exposure Limit                                                                | <b>RPE</b> - Respiratory Protective Equipment                                                                                |
| <b>DNEL</b> - Derived No Effect Level                                                                | <b>NOEC</b> - No Observed Effect Concentration                                                                               |
| <b>POW</b> - Partition coefficient Octanol:Water                                                     | <b>BCF</b> - Bioconcentration factor                                                                                         |
| <b>vPvB</b> - very Persistent, very Bioaccumulative                                                  | <b>PBT</b> - Persistent, Bioaccumulative, Toxic                                                                              |
| <b>VOC</b> - (Volatile Organic Compound)                                                             |                                                                                                                              |

#### **Key literature references and sources for data**

HSNO classifications provided in the New Zealand Chemical Classification Information Database (CCID).  
<https://echa.europa.eu/information-on-chemicals>  
Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS  
EPA Guide to classifying hazardous substances in New Zealand  
EPA - Assigning a product to an existing HSNO approval guide

#### **Training Advice**

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.  
Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.  
First aid for chemical exposure, including the use of eye wash and safety showers.  
Chemical incident response training.

|                         |                 |
|-------------------------|-----------------|
| <b>Revision Date</b>    | 22-Mar-2023     |
| <b>Revision Summary</b> | Initial Release |

#### **Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

## End of Safety Data Sheet